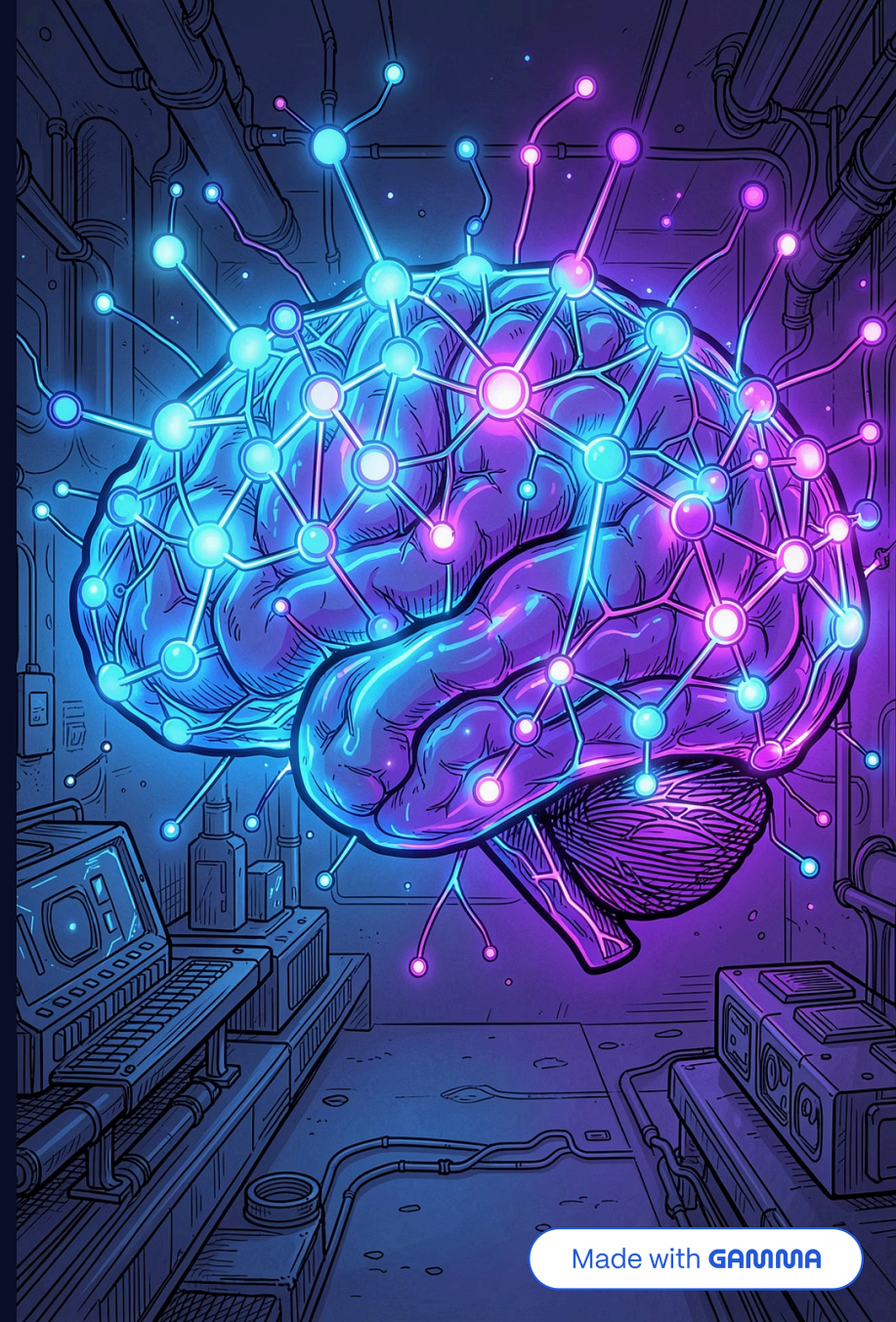


UNIT II – ARTIFICIAL INTELLIGENCE

Intelligent Agents

Goals and Agents in Artificial Intelligence – exploring how machines perceive, decide, and act.



By the End of This Lecture

01

Define an Agent

Understand what an agent is and its core components.

02

Explain Agent Structure

Describe how sensors, programs, and actuators work together.

03

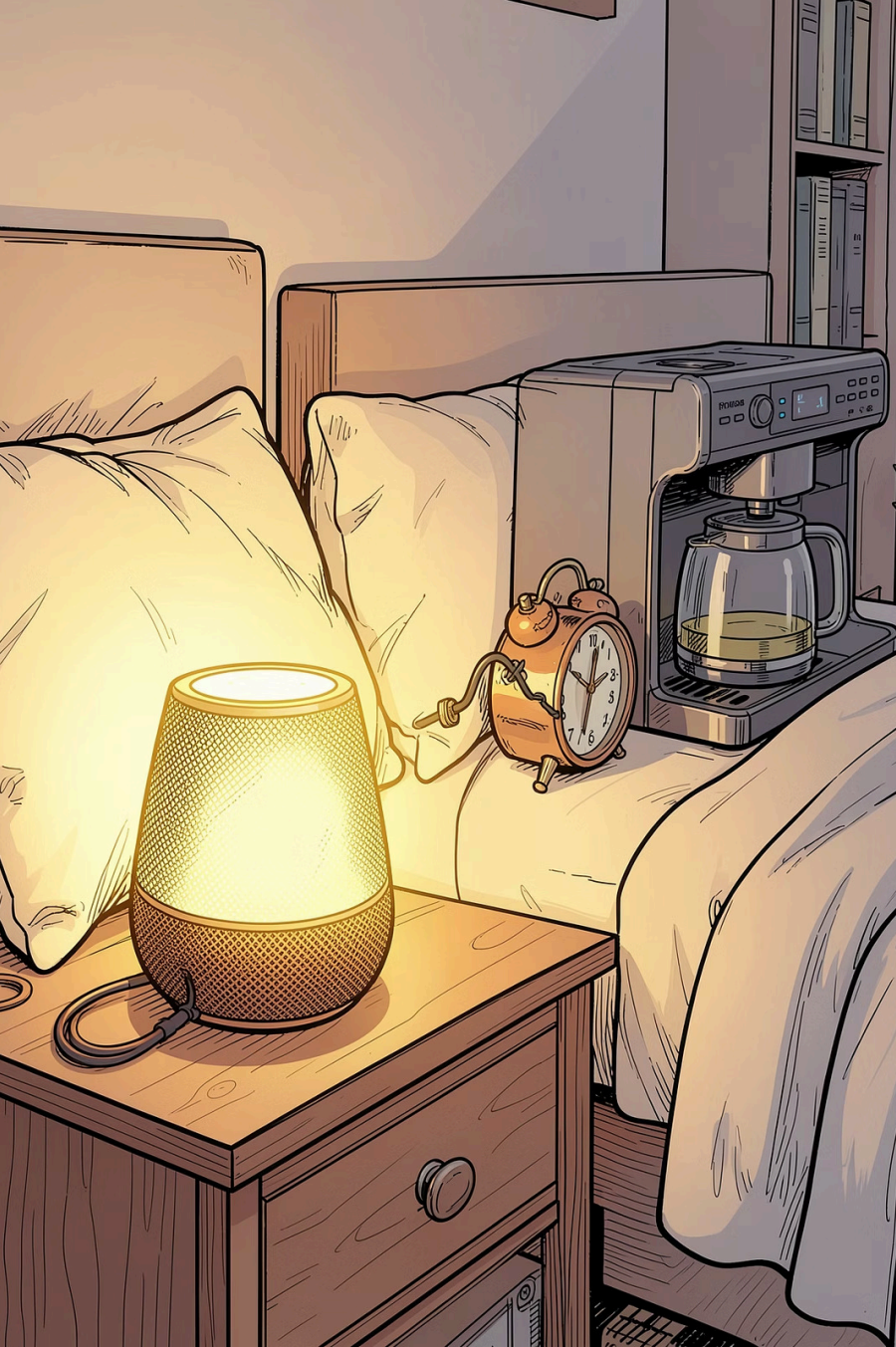
Identify Agent Types

Distinguish between the five types of intelligent agents.

04

Apply the PEAS Framework

Design and analyze agents using Performance, Environment, Actuators, and Sensors.



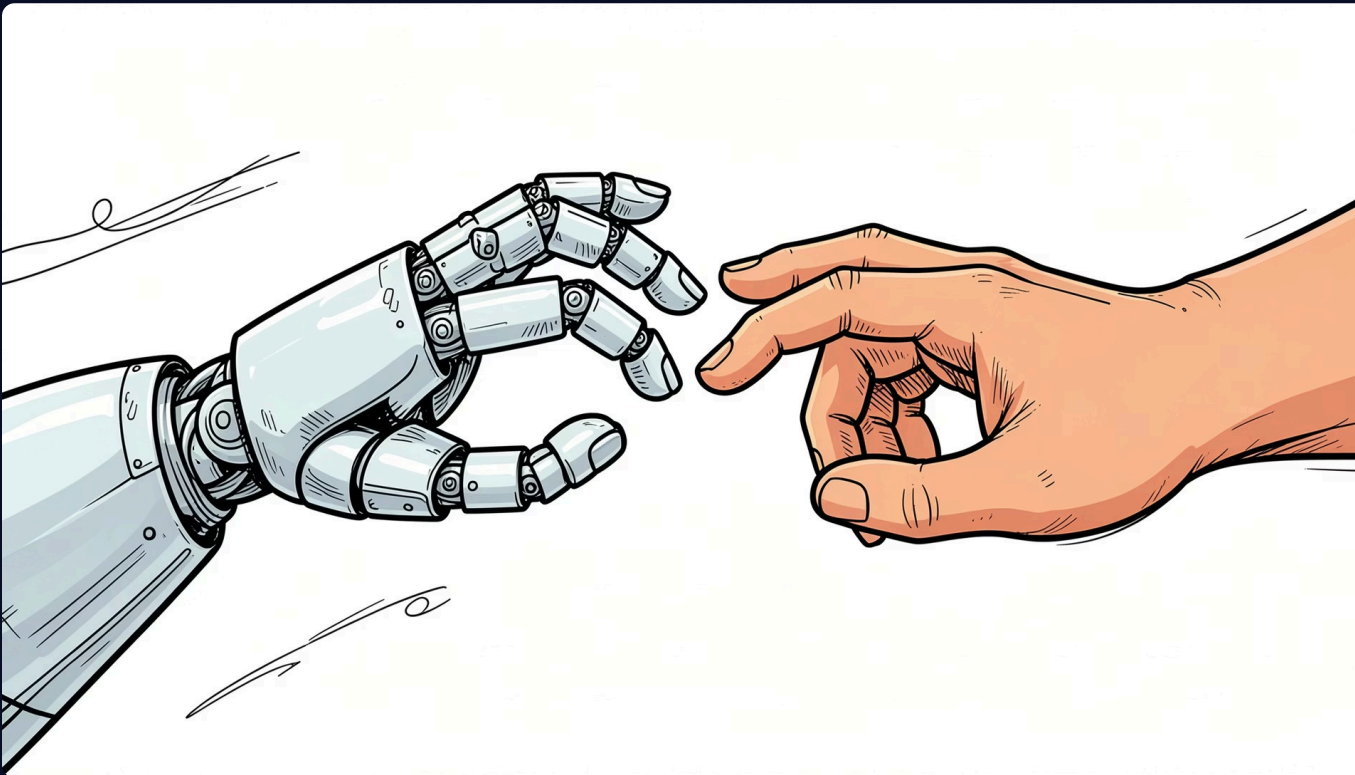
INTRODUCTION

Imagine This...

You say: "**Alexa, turn off the alarm.**" Instantly – the alarm stops, lights turn on, weather is announced, and the coffee machine starts.

- ❑ Did someone manually trigger all of this? No! An **Intelligent Agent** perceived your command and made those decisions autonomously.

What is an Agent?



Definition

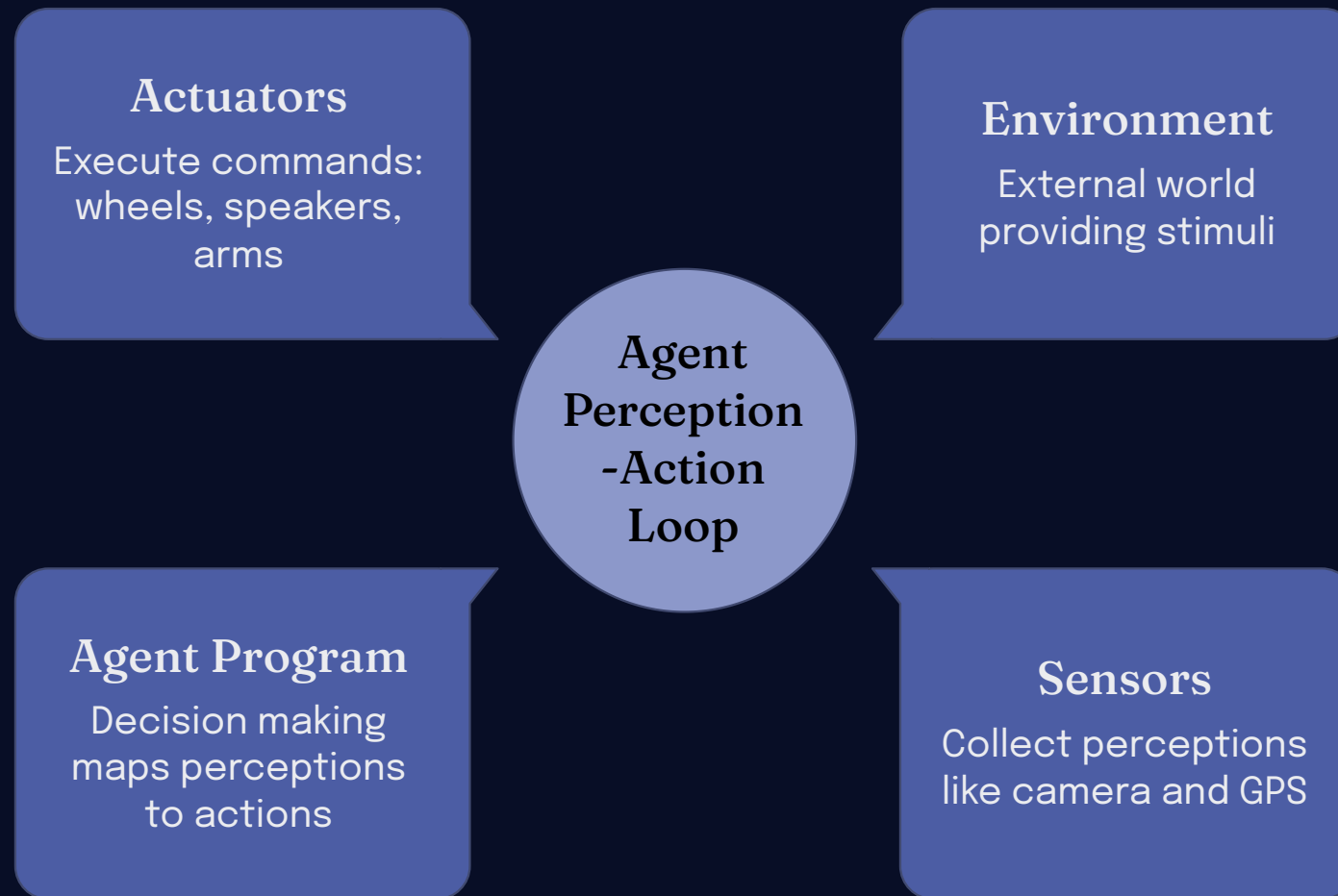
An **Agent** is anything that **observes** its environment, **makes decisions**, and **performs actions** to achieve a goal.

☐ **Agent = Perception + Decision + Action**

Real-Life Examples

Agent	Environment	Action
Human	Eyes, ears	Walk, speak
Robot	Camera, sensors	Move
Alexa	Voice	Respond
Self-driving car	Road	Drive

Components & Structure of an Agent



Every agent follows the cycle: **Observe** → **Think** → **Act**. Sensors collect information (camera, microphone, GPS); actuators perform actions (speaker, wheels, robot arms). The **Agent Function** maps perceptions to actions theoretically, while the **Agent Program** is the actual software implementation running on physical **Architecture** (hardware like a robot body, computer, or smartphone).

What Makes an Agent Intelligent?



Autonomous

Operates without direct human control.



Rational

Chooses the best action given available information.



Reactive

Responds promptly to changes in the environment.



Learns

Improves over time using experience and feedback.



Adaptable

Adjusts behaviour when the environment changes.

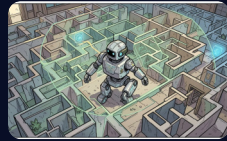
❏ **Rational ≠ Perfect.** A rational agent makes the *best possible* decision with the information it has – like Google Maps choosing the faster route (30 min vs 45 min).

Five Types of Intelligent Agents



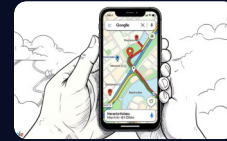
1. Simple Reflex

Acts on **current situation only**. Rule: IF condition → THEN action. *Example: Automatic door.*



2. Model-Based

Uses **internal memory** of past states. *Example: Robot remembering a wall behind it.*



3. Goal-Based

Works toward a **specific goal**. *Example: Google Maps finding the best route.*



4. Utility-Based

Chooses the **best option among many** by weighing trade-offs. *Example: Swiggy balancing speed, cost, and rating.*

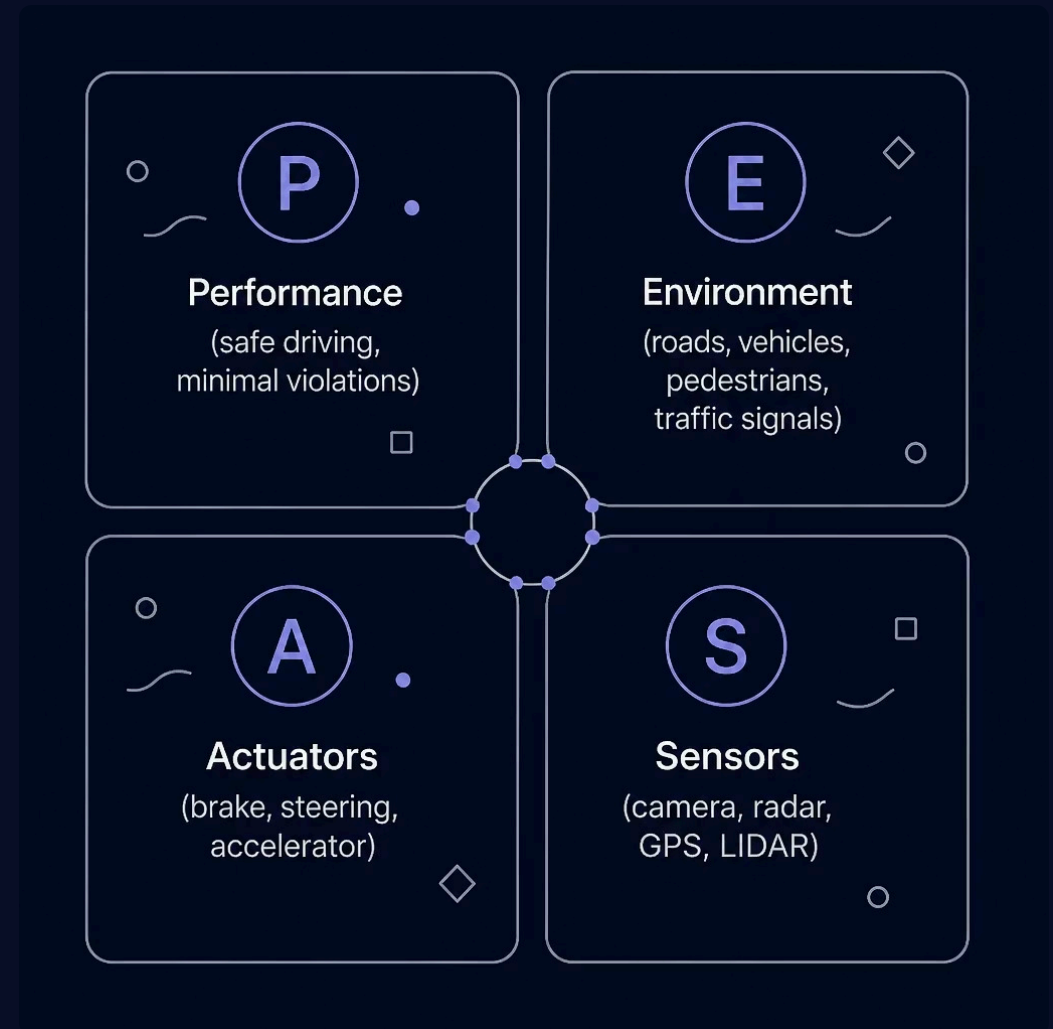


5. Learning Agent

Improves over time using experience. *Example: Netflix, Spotify recommendations.*

The PEAS Framework

PEAS is a structured method to **design and describe** any intelligent agent by specifying four key elements.



P — Performance

How success is measured.

E — Environment

Where the agent operates.

A — Actuators

How the agent acts.

S — Sensors

How the agent perceives.

Advantages, Limitations & Activity

✓ Advantages

- Faster, more accurate decisions
- Automation reduces human effort
- Operates 24×7 without fatigue
- Learns and improves continuously

⚠ Limitations

- Expensive to develop and deploy
- Requires high-quality training data
- Cannot interpret human emotions well
- Security and privacy risks
- Wrong decisions if data is flawed

❓ **Classroom Activity (5 min):** For each agent below, identify its Sensors and Actuators: **ATM · Drone · ChatGPT · Washing Machine**

Summary & Assignment

Key Takeaways

- **Agent** = Perception + Decision + Action
- **Structure:** Sensors → Program → Actuators
- **5 Types:** Reflex, Model, Goal, Utility, Learning
- **PEAS** designs and evaluates agents

Assignment

1. Define an Intelligent Agent with a real-life example.
2. Draw and explain the structure of an agent.
3. Differentiate Agent Function and Agent Program.
4. Explain all five types of intelligent agents with examples.
5. Design a PEAS framework for a Smart Traffic Signal.